AMRITA VISHWA VIDYAPEETHAM
DEEMED TO BE UNIVERSITY UNDER SECTION 3 OF UGC ACT, 1956
COIMBATORE CAMPUS

INSTITUTION'S INNOVATION COUNCIL
(Ministry of Education Initiative)

GENESIS

Genesis Ideathon: Final Presentation

Team ID: *GEN26-A0H-I204*

Team Name: *Byteblaze*

Domain: *Intelligent Systems Digital Futures*

Amrita Vishwa Vidyapeetham, Coimbatore Campus

Date: 11th October 2026

Presentation Outline

1. Problem Statement
2. Problem Understanding
3. Existing Gaps & Challenges
4. Our Proposed Solution
5. Technical Approach
6. Innovation & Uniqueness
7. Feasibility & Scalability
8. Cost for Product
9. Use Cases
10. Impacts and Benefits
11. Sustainable Development Goals
12. Individual Member Contribution

Problem Statement

Creating a tool that identifies suspicious links, phishing messages, fake websites and unsafe emails before students interact with them.

Problem Understanding

- Students, teachers, educational institutions, and organizations that receive suspicious links, fake emails, and fraudulent messages are affected.
- Cybercriminals create fake websites, phishing links, and misleading messages to impersonate trusted sources and steal sensitive information.
- It can lead to data theft, financial loss, privacy violations, and reduced trust in online services.

Existing Gaps & Challenges

- Use of friendly services to obscure phishing URLs.
- High similarity between the HTML structures of phishing and benign pages.
- Due to the growth of technology, security threats to systems and networks are relentlessly inventive.

Our Proposed Solution

- AI/ML based cybersecurity tool with python flask backend.
- The tool analyzes URLs, message content, and suspicious keywords to identify potential threats and displays warnings with reasons.
- It helps students stay safe online, protect personal information, avoid scams, and improve cybersecurity awareness.

Technical Approach

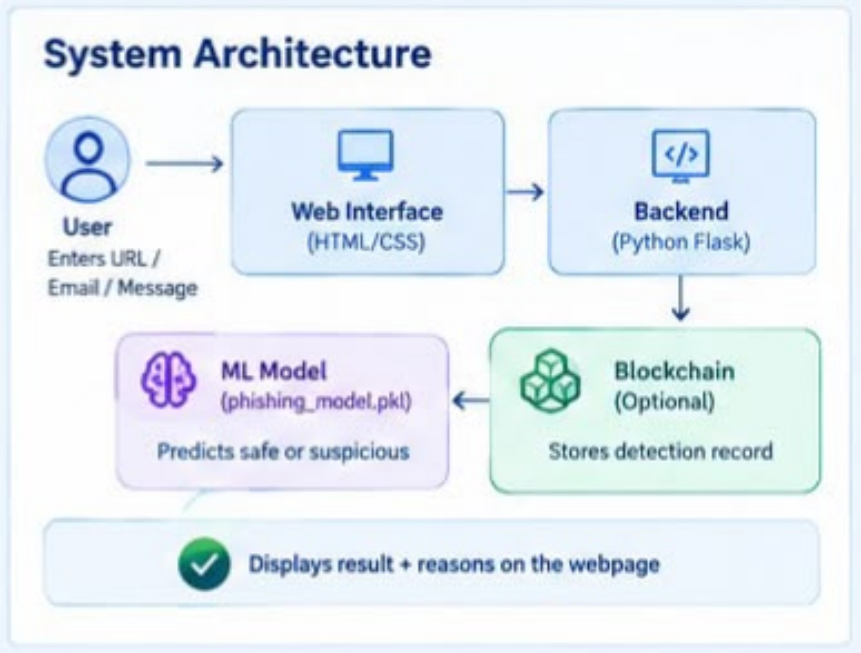


Figure: Enter Caption

Tech Stack

Frontend: CSS,HTML

Database: MySQL

Backend: Pythonflask, MySQL

AI/ML, APIs, or Hardware: AI/ML

Innovation & Uniqueness

- Our tool detects suspicious links, phishing messages, fake websites, and unsafe emails in one place while explaining the reasons for each warning.
- It helps students identify online threats quickly through simple analysis and easy-to-understand safety alerts.
- Its main differentiator is its student-friendly approach, combining threat detection with clear explanations to improve cybersecurity awareness.

Feasibility & Scalability

- **The tool can be developed using Python, basic cybersecurity knowledge, URL analysis techniques, and a simple web interface.**
- **The tool can be improved by adding smart learning technology, updated lists of dangerous links, and browser extensions to protect more people online.**
- **Limited detection accuracy, evolving cyber threats, and privacy concerns are key risks that can be reduced through regular updates, testing, and secure data handling.**

Cost for Product (Optional)

- Development or hardware cost: The tool can be developed at a low cost using Python and free software tools. A basic computer and an internet connection are sufficient, so no special hardware is required.
- Hosting or maintenance cost: The tool can be hosted using free hosting platforms, with minimal maintenance costs. Regular software updates and security checks are needed to keep it working safely and effectively.
- Estimated total: 0–2,000 for a basic prototype using free software tools and platforms, excluding any existing computer or internet costs.

Use Cases (compulsorily 3 use cases must be there)

- Use case 1: A student receives a scholarship link and uses our tool to check whether it is safe before clicking it.
- Use case 2: A student receives a fake internship email, and the tool identifies suspicious words and warns about possible fraud.
- Use case 3: A student checks an unfamiliar placement website, and the tool analyzes its URL and alerts them to potential risks.

Impacts and Benefits

- Helps students identify phishing links, fake websites, and unsafe emails to protect their personal information and stay safe online.
- Improves cybersecurity awareness, reduces the risk of financial loss, and encourages safer use of digital services.
- Aims to detect suspicious links and messages, provide clear safety warnings, and improve students' ability to recognize online threats.

Sustainable Development Goals (SDGs)

SDG Number	Target Number	Goal Name	Connection
SDG 4	Target 4.4	Quality Education	Our tool improves students' digital skills and cybersecurity awareness, helping them use online resources safely and confidently.
SDG 9	Target 9.1	Industry, Innovation and Infrastructure	Our tool uses technology to identify online threats and supports the development of safer and more reliable digital systems.

Individual Member Contribution

Member	Contribution or responsibility
Harini S	PPT ,Research, Submission
Nikitha Varshini Parthiban	Research, Chart work ,Github respository
Lasyapriya C	Research, Chart work
Riduvarshini	Research, Chart work

Yukti ID

IR2026-1051104